

Payments Architect — Express Lane Items to Land — Q2 2026 (PAY-4026)

 [PAY-4032] payment-mgmt: move ACH redeposit-mint off legacy API (Pattern D') Created: 28/Jun/26 Updated: 13/Jul/26

Status:	In Stage
Project:	Payments
Components:	None
Affects versions:	None
Fix versions:	R9.6
Parent:	Payments Architect — Express Lane Items to Land — Q2 2026

Type:	Story	Priority:	2: Important
Reporter:	Jason Thomas	Assignee:	Brett Roy
Resolution:	Unresolved		
Labels:	express-lane		
Σ Remaining Estimate:	0 minutes	Remaining Estimate:	Not Specified
Σ Time Spent:	1 day, 5 hours, 15 minutes	Time Spent:	Not Specified
Σ Original Estimate:	Not Specified	Original estimate:	Not Specified

Attachments:	 PAY-4032-test-runbook.md  PAY-4032-ach-redeposit.postman_collection.json				
Sub-tasks:	Key	Summary	Type	Status	Assignee
	PAY-4085	DEV - payment-mgmt: move ACH redeposi...	Sub-task	Done	Brett Roy
	PAY-4099	QA: Create Test Cases	Sub-task	In Progress	Christopher Ariel Valenzuela Medina
	PAY-4100	QA: Test in Stage	Sub-task	In Progress	Christopher Ariel Valenzuela Medina
Sprint:	PAY-Sprint 17				
QA Required:	Yes				
Grooming Status:	7-Groomed				
Pod:	PAY				

Description

The ACH-return redeposit-mint leg still runs in legacy payments-api (AchReturnReceivedEventHandler), while Payment Management already consumes Payment.Return.Received over Service Bus (post-#212) and transitions the original debit to RETURNED without minting the retry.

Add a retry-mint branch to PaymentReturnReceivedHandler.ProcessDebitReturn gated on IsRetryable: mint a child PaymentTransaction with parent/root lineage and RetryAttempt+1, submit via a new IWorldpayService.CreateEcheckRedepositAsync (WP-Wrapper redeposit endpoint, re-presenting by ProcessorTransactionId), and publish Payment.Debit.StatusChanged in the existing single Unit of Work. Includes implementing the WP-Wrapper CreateEcheckRedeposit endpoint (currently a NotImplementedException stub) and a strangler retirement of the legacy handler + its EGN subscription. Idempotency via the entity transition guard + unique filtered index on (RootTransactionId, RetryAttempt); behind flag PaymentManagement:AchRetry:Enabled. Pattern D' locked 2026-05-31.

Full analysis + solution: `_bmad-output/products/payments/04_ExpresseLane/enhancement/ach-retry-from-legacy-payments-to-payment-management/`

Comments

Comment by Ana Paula [08/Jul/26]
Swetha Bayya Shikha Atal Andy Holtsclaw Brett Roy need to discuss if this is QA required or not.
Comment by Swetha Bayya [08/Jul/26]
Ana Oliva (contractor)
QA validation is recommended because this story moves ACH redeposit/retry logic from the legacy Payments API to Payment Management, the expected business behavior remain the same, this can be covered as part of end-to-end ACH regression testing for the release, instead of a separate QA cycle. Regression should include a returned ACH transaction followed by redeposit/retry processing to confirm the flow still works correctly after migration.
Brett Roy

Comment by [Brett Roy](#) [09/Jul/26]

This required a bit of back and forth on the design and feedback from Jason - bumping to SP = 3

Comment by [Ana Paula](#) [09/Jul/26]

[Shikha Atal Swetha Bayya](#) since this is now QA required can you please add Acceptance criteria for test design. thanks!

Comment by [Shikha Atal](#) [09/Jul/26]

[Andy Holtsclaw](#) since Swetha is out today, will you be able to add Acceptance Criteria as requested by QA.

Comment by [Brett Roy](#) [13/Jul/26]

Test artifacts (runbook + Postman collection)

- Attached to this ticket:
- **PAY-4032-test-runbook.md** — the test runbook (chain map, coverage analysis, Part A procedure, parked Part B + DEV synthetic-smoke steps, DEV run results).
 - **PAY-4032-ach-redeposit.postman_collection.json** — Postman companion. Import into Postman; set the collection variables (wpWrapperBaseUrl, clientId/clientSecret, tokenScope=WorldpayWrapperAPI.Full, accountId, subMerchantId) for your target env.

What they cover

- *Part A (runnable today)*: the WorldpayWrapper redeposit HTTP boundary (POST /api/v1/echeck/redeposits) — the one real HTTP surface in the retry chain. Executed live on DEV (real Worldpay Pre-Live): token OK; seed sale 000; redeposit of a fresh sale returns 367 ("deposit has not been returned for NSF") = correct Vantiv behavior + faithful mapper round-trip; input-validation guards pass.
- *Event handling* (consume Payment.Return.Received → mint retry child → emit Payment.Retry.Submitted → payments-func mints the Charge) is covered by the automated unit + integration tests that shipped in wellfit-payments PR #319. There is no runnable manual/E2E test of the event chain today.
- *Part B (E2E, parked)*: full-chain synthetic-inject procedure documented, plus a near-term DEV-only synthetic smoke (needs only the config setting PaymentManagement_AchRetry_Enabled=true in DEV + legacy consumer kept quiet).

Two open gaps flagged (no tickets exist):

1. B3 inbound DTO field-name mismatch — ach-returns emits OriginalPaymentId; PM binds paymentId → dead-letter. Prerequisite for flag-ON with real traffic.
2. Slice-6 legacy retirement — legacy payments-api still consumes the return and still mints the redeposit; the PM flag must stay OFF until it is retired.

Comment by [Brett Roy](#) [13/Jul/26]

[Radhika Kandoori Christopher Ariel Valenzuela Medina Ana Oliva \(contractor\)](#)

Added runbook for testing, it is a limited set of tests as there are some upcoming changes that will allow for further tests!

[Swetha Bayya Shikha Atal](#)